

# Automated Table Tennis Ball Sorter

4 classifications38mm & 40mm diameterWhite & Orange colourCustom PCB · 3D-printed · ATmega328P-PU

## THE PROBLEM

Since the 2000 ITTF regulation change, two incompatible standards coexist — **38mm celluloid** (legacy) and **40mm ABS plastic** (current). Visually indistinguishable to the naked eye, manual sorting introduces consistent classification errors and significant labour overhead in training and equipment management contexts.

## KEY PERFORMANCE TARGETS

100%

ACCURACY

Diameter sorting  
(38 vs 40mm)

≥ 92%

ACCURACY

Colour detection  
(white vs orange)

≤ 5s

PER BALL

Full cycle:  
feed → detect → divert

≤ 4

MINUTES

Full 40-ball  
batch runtime

≥ 98%

JAM-FREE

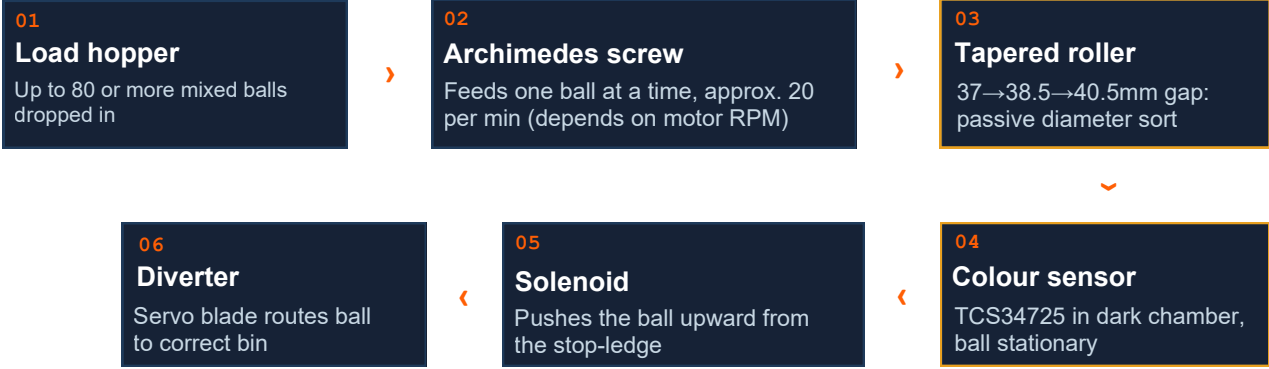
40-ball continuous  
operation

< 10ms

LATENCY

IR trigger →  
solenoid actuation

## SORTING SEQUENCE — HOW IT WORKS



## SUBSYSTEM BREAKDOWN

Feeding System

Archimedes screw · IR sensor

➤ 3D-printed PLA screw — 42mm flight spacing, 200mm length

➤ JGY-370 12V DC gear motor — 30–60 RPM, ≥ 5 kg·cm torque

➤ FC-51 IR sensor — detects empty hopper, stops motor automatically

➤ Inclined hopper base eliminates dead zones; all balls self-feed to screw inlet

Sorting System

Tapered roller · colour station · diverter

➤ Tapered rollers — passive 37→38.5→40.5mm gap, 300mm length, PLA printed

➤ TCS34725 colour sensor in sealed dark chamber; ball held stationary on stop ledge

➤ 12V solenoid (10–15mm stroke) lifts ball after colour reading is complete

➤ MG90S servo drives single-blade diverter 60° to route ball to one of four bins

Power System

12V input · LM2596 buck converter · 5V logic rail

➤ 12V 3A wall adapter → LM2596 buck converter → 5V rail for MCU, sensors, servos

➤ Isolated power paths — high-current (motor, solenoids) separated from logic rail

➤ 2A slow-blow fuse + 1N4007 reverse polarity protection diode on main input

➤ Resettable fuse (TLC-MSMD200) on motor path for overcurrent protection

Control System

Custom KiCAD PCB · ATmega328P

➤ ATmega328P-PU on custom KiCAD PCB — Arduino-bootloaded, 16 MHz crystal

➤ TB6612FNG H-bridge for DC motor · IRLZ44N MOSFETs for solenoid switching

➤ TCA9548A PC multiplexer addresses both TCS34725 colour sensors independently

➤ State machine: IDLE → FEEDING → DETECTING → DIVERTING → DONE

## KEY ENGINEERING DECISIONS

Mechanical vs sensor-based sizing

Tapered rollers over sensor-based diameter detection

Passive mechanical sorting requires zero calibration, zero electronics, and produces no false positives from sensor noise or lighting variation. The gap geometry itself is the classifier — no software logic required.

Stationary vs in-motion reading

Stop-ledge colour detection over inline sensing

The TCS34725 is highly sensitive to motion and ambient light. Holding the ball stationary on a stop ledge inside a sealed dark chamber before reading eliminates both error sources, reliably achieving the ≥90% accuracy target.